

Installation of an Ubuntu Web Server (Apache)

Objective:

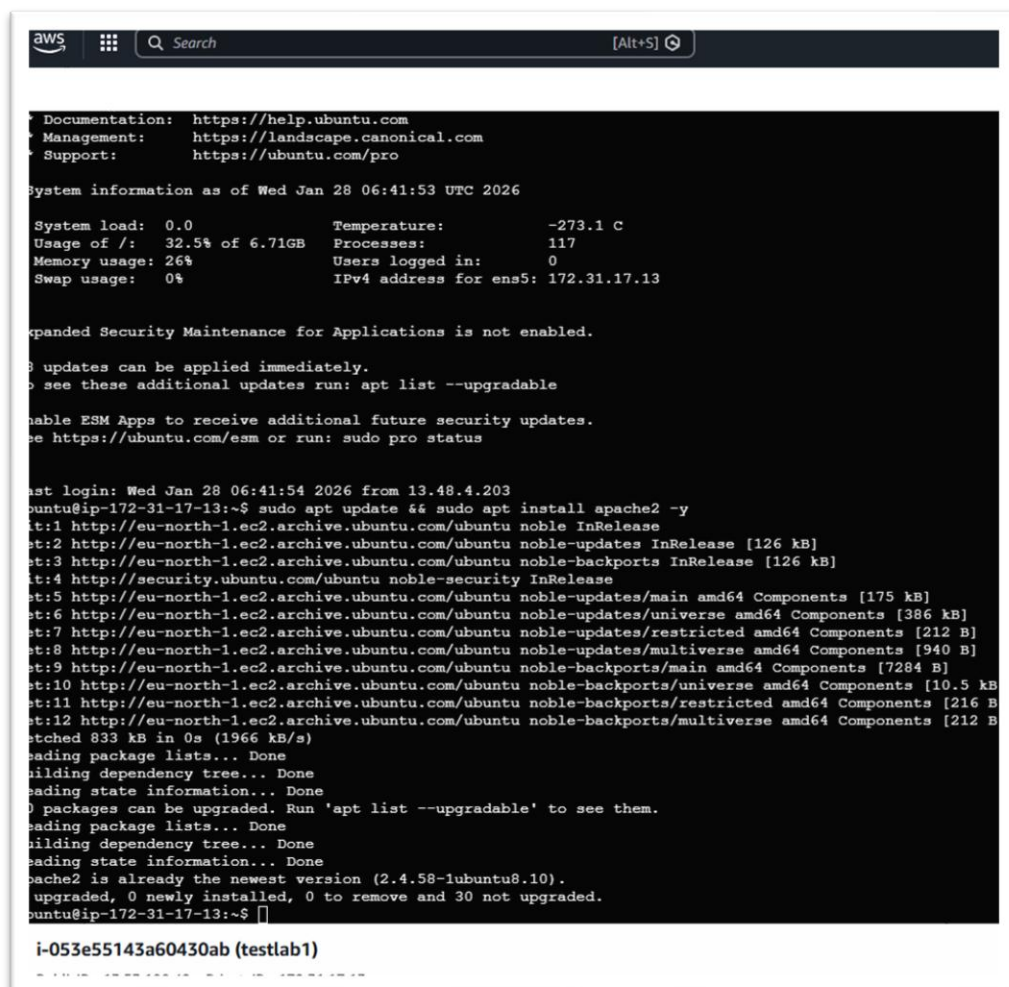
To install and configure an Apache web server on an Ubuntu system and verify that it is accessible via a web browser.

Step 1: Update the System and Install Apache

The system package index was updated, and the Apache web server was installed using the Advanced Package Tool (APT).

Bash Script:

```
sudo apt update && sudo apt install apache2 -y
```



```
aws [Search] [Alt+S]
Documentation: https://help.ubuntu.com
Management: https://landscape.canonical.com
Support: https://ubuntu.com/pro

System information as of Wed Jan 28 06:41:53 UTC 2026

System load: 0.0 Temperature: -273.1 C
Usage of /: 32.5% of 6.71GB Processes: 117
Memory usage: 26% Users logged in: 0
Swap usage: 0% IPv4 address for ens5: 172.31.17.13

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Jan 28 06:41:54 2026 from 13.48.4.203
ubuntu@ip-172-31-17-13:~$ sudo apt update && sudo apt install apache2 -y
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Hit:6 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Hit:7 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Hit:8 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Hit:9 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7284 B]
Hit:10 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Hit:11 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Hit:12 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 833 kB in 0s (1966 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu8.10).
0 upgraded, 0 newly installed, 0 to remove and 30 not upgraded.
ubuntu@ip-172-31-17-13:~$
```

This command ensures the latest package information is retrieved and installs Apache without requiring manual confirmation.

Step 2: Verify Apache Service Status

After installation, the Apache service was checked to confirm it was running successfully.

Bash Script:

sudo systemctl status apache2

```
aws [Search] [Alt+S]

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Jan 28 06:41:54 2026 from 13.48.4.203
ubuntu@ip-172-31-17-13:~$ sudo apt update && sudo apt install apache2 -y
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:5 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:6 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:7 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:8 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:9 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7284 B]
Get:10 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:11 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:12 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 833 kB in 0s (1966 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
30 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu8.10).
0 upgraded, 0 newly installed, 0 to remove and 30 not upgraded.
ubuntu@ip-172-31-17-13:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-01-28 06:10:28 UTC; 35min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 16435 (apache2)
    Tasks: 55 (limit: 1008)
   Memory: 5.4M (peak: 5.6M)
      CPU: 141ms
   CGroup: /system.slice/apache2.service
           └─16435 /usr/sbin/apache2 -k start
           └─16437 /usr/sbin/apache2 -k start
           └─16438 /usr/sbin/apache2 -k start

Jan 28 06:10:28 ip-172-31-17-13 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jan 28 06:10:28 ip-172-31-17-13 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-17-13:~$
```

i-053e55143a60430ab (testlab1)

PublicIPs: 13.53.190.42 PrivateIPs: 172.31.17.13

Step 3: Configure Firewall Rules (If enabled)

The Uncomplicated Firewall (UFW) was checked and configured to allow HTTP traffic on port 80.

Bash Script:

```
sudo ufw allow 'Apache'
```

```
sudo ufw reload
```

```
ubuntu@ip-172-31-17-13:~$ sudo ufw allow Apache
Rules updated
Rules updated (v6)
ubuntu@ip-172-31-17-13:~$ sudo ufw reload
Firewall not enabled (skipping reload)
ubuntu@ip-172-31-17-13:~$
```

i-053e55143a60430ab (testlab1)

PublicIP: 13 53 190 42 PrivateIP: 172 31 17 13

This ensures that external clients can access the web server.

Step 4: Test Web Server Accessibility

The server's IP address was identified and used to test Apache in a web browser.

Bash Script:

```
hostname -I
```

```
ubuntu@ip-172-31-17-13:~$ hostname -I
172.31.17.13
ubuntu@ip-172-31-17-13:~$
```

i-053e55143a60430ab (testlab1)

PublicIP: 13 53 190 42 PrivateIP: 172 31 17 13

Navigating to <http://172.31.17.13> displayed the **Apache2 Ubuntu Default Page**, confirming successful deployment.

```
DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"
html xmlns="http://www.w3.org/1999/xhtml">
<!--
  Modified from the Debian original for Ubuntu
  Last updated: 2022-03-22
  See: https://launchpad.net/bugs/1966004
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
<title>Apache2 Ubuntu Default Page: It works</title>
<style type="text/css" media="screen">
* {
margin: 0px 0px 0px 0px;
padding: 0px 0px 0px 0px;
}
body, html {
padding: 3px 3px 3px 3px;
background-color: #D8DBE2;

```

i-053e55143a60430ab (testlab1)

PublicIP: 13 53 190 42 PrivateIP: 172 31 17 13

Step 5: Identify the Web Root Directory

Apache's default document root was verified.

Bash Script:

```
cd /var/www/html
```

```
ls
```



```
buntu@ip-172-31-17-13:~$ cd /var/www/html
buntu@ip-172-31-17-13:/var/www/html$ ls
index.html
buntu@ip-172-31-17-13:/var/www/html$
```

i-053e55143a60430ab (testlab1)

PublicIPs: 13.53.190.42 PrivateIPs: 172.31.17.13

The default **index.html** file confirmed the web server's content location

1. Apache running? **sudo systemctl status apache2**

- ## 2. Listening on port 80? `ss -tulnp | grep apache2`

DokId: 12 52 100 42 DrustId: 172 21 17 12

- ## 2. Listening on port 80? `ss -tulnp | grep apache2`

DocId: 12 52 100 42 DocId: 172 21 17 12

- #### 4. Testing locally on the server itself **curl** <http://localhost>

PublicID: 13 53 100 12 PrivateID: 172 31 17 12

w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd

